



SELF-SERVICE PROVISIONER BOT

**AI-Powered Infrastructure Provisioning
and GitOps Automation**

Instant Environment Setup in Seconds

From Natural Language Requests → Production-Ready Infrastructure

Team NextGen Solutions – Team-15

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PROBLEM STATEMENT

Infrastructure provisioning in modern organizations is primarily a **manual, time-consuming**, and **error-prone** process. Teams rely on disconnected tools and repetitive steps, leading to **delays, inconsistencies**, and **governance risks**.

KEY CHALLENGES



1. MANUAL & TIME-CONSUMING

Engineers spend hours creating infrastructure configurations, writing YAML/Terraform files, and setting up environments.



2. ERROR-PRONE PROCESS

Manual configurations often lead to syntax errors, misconfigurations, and inconsistent environments across teams.



3. DISCONNECTED WORKFLOWS

Multiple tools for tickets, code, CI/CD, and notifications cause context switching and lack of end-to-end visibility.



4. POLICY & COMPLIANCE RISKS

Hard to enforce organizational policies, naming conventions, and environment-specific guardrails consistently.



5. SLOW TURNAROUND TIME

From request to production, it takes days due to manual approvals, PR creation, and deployment delays.



6. POOR EXPERIENCE

Developers and stakeholders lack real-time updates and have no easy way to track request status.

IMPACT



WASTED TIME

High engineering effort spent on repetitive and low-value tasks.



INCONSISTENCY

Different environments and teams follow different processes.



SECURITY RISKS

Policy violations and misconfigurations can lead to compliance issues.



LOW PRODUCTIVITY

Developers are blocked and unable to focus on building core features.



There is a critical need for an intelligent, automated, and end-to-end solution that simplifies and accelerates infrastructure provisioning.



OUR MISSION

Build an AI-powered automation platform that converts natural language requests into production-ready infrastructure through validated, gitops-driven workflows.

SOLUTION OVERVIEW & EXPECTED OUTCOMES



KEY SOLUTION CAPABILITIES



AI-POWERED REQUEST UNDERSTANDING

Converts plain English requests into structured requirements



POLICY-DRIVEN VALIDATION

Ensures compliance with organizational standards and security policies



AUTOMATED INFRASTRUCTURE GENERATION

Generates Docker Compose and Terraform configurations automatically



GITOPS WORKFLOW AUTOMATION

Creates GitHub Issues and Pull Requests for complete traceability



CI/CD DEPLOYMENT AUTOMATION

Enables seamless deployment through GitHub Actions



REAL-TIME NOTIFICATIONS

Keeps developers and stakeholders informed throughout the process



EXPECTED OUTCOMES



REDUCED PROVISIONING TIME

from **hours** to **minutes**



IMPROVED DEPLOYMENT CONSISTENCY



ENHANCED SECURITY AND GOVERNANCE



INCREASED DEVELOPER PRODUCTIVITY

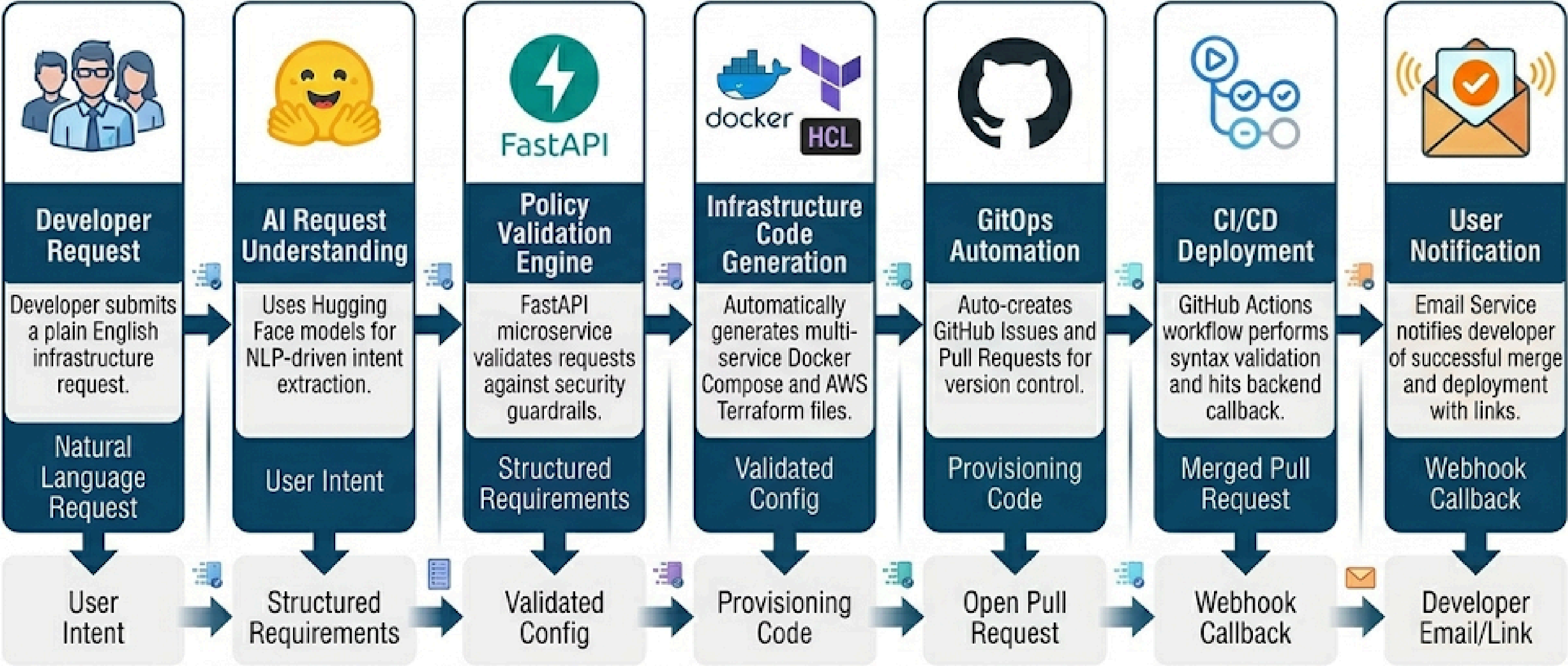


END-TO-END INFRASTRUCTURE AUTOMATION



FROM NATURAL LANGUAGE REQUESTS TO **PRODUCTION-READY INFRASTRUCTURE** IN MINUTES.

PROVISION COPILOT – END-TO-END INFRASTRUCTURE AUTOMATION PIPELINE



AUTOMATION TOOLS

AI-Powered Infrastructure Provisioning & GitOps Automation Platform

01

Tool 1 – AI Request Parser

Technology: Hugging Face (FLAN-T5)



- Understands natural language requests
- Extracts environment type
- Identifies required services
- Converts request into structured format

02

Tool 2 – Policy Validation Engine

Technology: Python Validation Logic + policy.yaml



- Checks allowed environments
- Validates service limits
- Enforces security policies
- Prevents invalid deployments

03

Tool 3 – Infrastructure Generator

Technology: Docker Compose & Terraform



- Generates docker-compose.yml
- Generates Terraform templates
- Creates production-ready configurations
- Supports cloud and container deployments

04

Tool 4 – GitOps Automation Engine

Technology: GitHub REST API



- Creates GitHub Issues
- Creates Pull Requests
- Maintains audit trail
- Tracks infrastructure changes

05

Tool 5 – CI/CD Deployment Engine

Technology: GitHub Actions



- Validates generated code
- Executes deployment workflow
- Triggers deployment webhook
- Updates deployment status

06

Tool 6 – Notification Service

Technology: Resend API / SMTP



- Sends submission notifications
- Sends deployment completion alerts
- Shares PR and Issue links
- Keeps stakeholders informed

TECHNOLOGIES USED



FastAPI



Hugging Face (FLAN-T5)



Docker Compose



Terraform



GitHub API



GitHub Actions



SQLite



Resend API



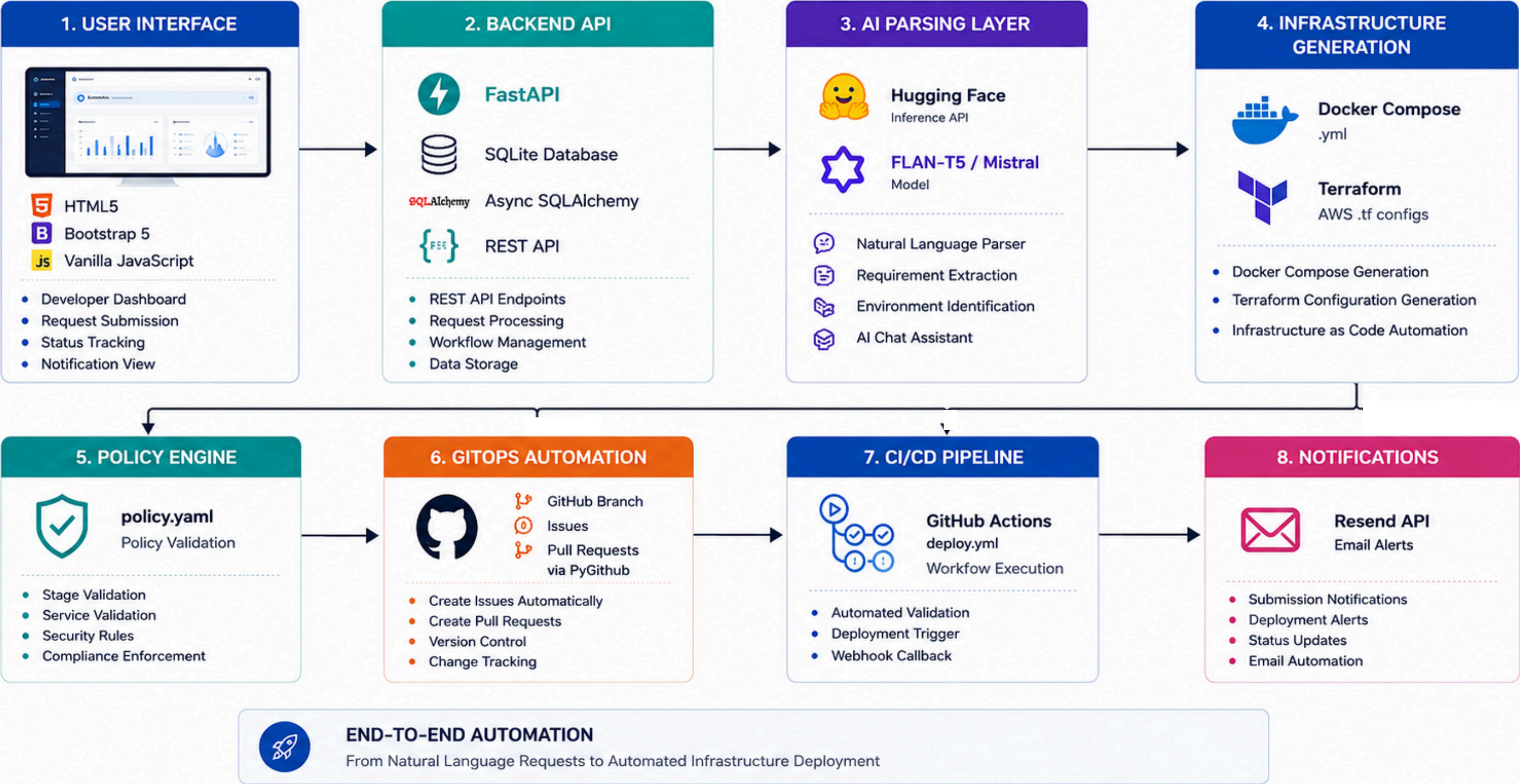
From Natural Language Requests to Production-Ready Infrastructure



TECHNOLOGY STACK

SELF-SERVICE PROVISIONER BOT

AI-Powered Infrastructure Provisioning & GitOps Automation Platform



WHAT MAKES PROVISION COPILOT UNIQUE

AI-Powered Infrastructure Provisioning & GitOps Automation Platform



1 AI-POWERED REQUEST UNDERSTANDING

- Understands natural language requests
- Extracts intent, environment & services
- Powered by advanced LLMs (Hugging Face)



2 INTELLIGENT VALIDATION ENGINE

- Policy-based validation before provisioning
- Ensures security, compliance & governance
- Prevents misconfigurations



3 AUTOMATED INFRASTRUCTURE GENERATION

- Generates Docker Compose or Terraform
- Consistent, reusable & production-ready code
- Reduces manual effort and errors



4 GITOPS-NATIVE WORKFLOW

- Automatically creates GitHub Issues & Pull Requests
- Full traceability & version control
- Collaborative & auditable process



5 AUTOMATED CI/CD DEPLOYMENT

- GitHub Actions for seamless deployments
- Consistent, repeatable & reliable delivery
- Environments ready in minutes



6 SMART NOTIFICATIONS

- Real-time email updates via Resend API
- Keeps teams informed at every stage
- Improves transparency & trust



PROVISION COPILOT – INTELLIGENT AUTOMATION. SECURE BY DESIGN. DELIVERED WITH SPEED.

Transforming natural language requests into production-ready infrastructure through AI, GitOps, and DevOps automation.



ADVANTAGES OF PROVISION COPILOT



1. FASTER INFRASTRUCTURE PROVISIONING

- ✓ Reduces setup time from hours or days to minutes
- ✓ Eliminates repetitive manual tasks



2. AI-POWERED AUTOMATION

- ✓ Converts natural language requests into infrastructure code
- ✓ Simplifies environment creation for developers



3. IMPROVED SECURITY & COMPLIANCE

- ✓ Validates requests against organizational policies
- ✓ Prevents unauthorized or misconfigured deployments



4. REDUCED HUMAN ERRORS

- ✓ Automated code generation ensures consistency
- ✓ Minimizes configuration mistakes



5. GITOPS-BASED WORKFLOW

- ✓ Automatic GitHub Issue and Pull Request creation
- ✓ Complete audit trail and version control



6. INCREASED DEVELOPER PRODUCTIVITY

- ✓ Developers focus on coding instead of infrastructure setup
- ✓ Faster project delivery



7. REAL-TIME NOTIFICATIONS

- ✓ Instant updates on request submission and deployment status
- ✓ Better communication among stakeholders



8. SCALABLE & FLEXIBLE

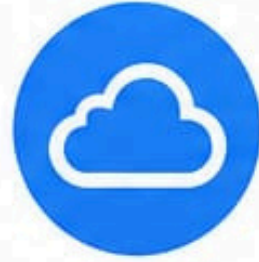
- ✓ Supports Docker Compose and Terraform
- ✓ Easily adaptable to different deployment environments

FUTURE ENHANCEMENTS



KUBERNETES SUPPORT

- ✓ Generate Kubernetes YAML manifests automatically
- ✓ Support container orchestration at scale



MULTI-CLOUD INTEGRATION

- ✓ Add support for Azure and Google Cloud Platform
- ✓ Enable cross-cloud deployments



ROLE-BASED ACCESS CONTROL (RBAC)

- ✓ Different access levels for developers, reviewers, and administrators
- ✓ Improved security management



COST ESTIMATION ENGINE

- ✓ Predict infrastructure costs before deployment
- ✓ Help organizations optimize cloud spending



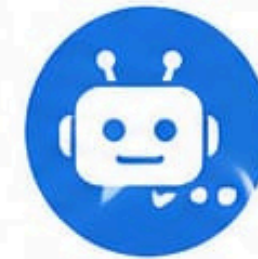
ADVANCED MONITORING DASHBOARD

- ✓ Real-time infrastructure health monitoring
- ✓ Performance analytics and reporting



ADVANCED AI RECOMMENDATIONS

- ✓ Suggest optimal infrastructure configurations
- ✓ Recommend resource scaling strategies



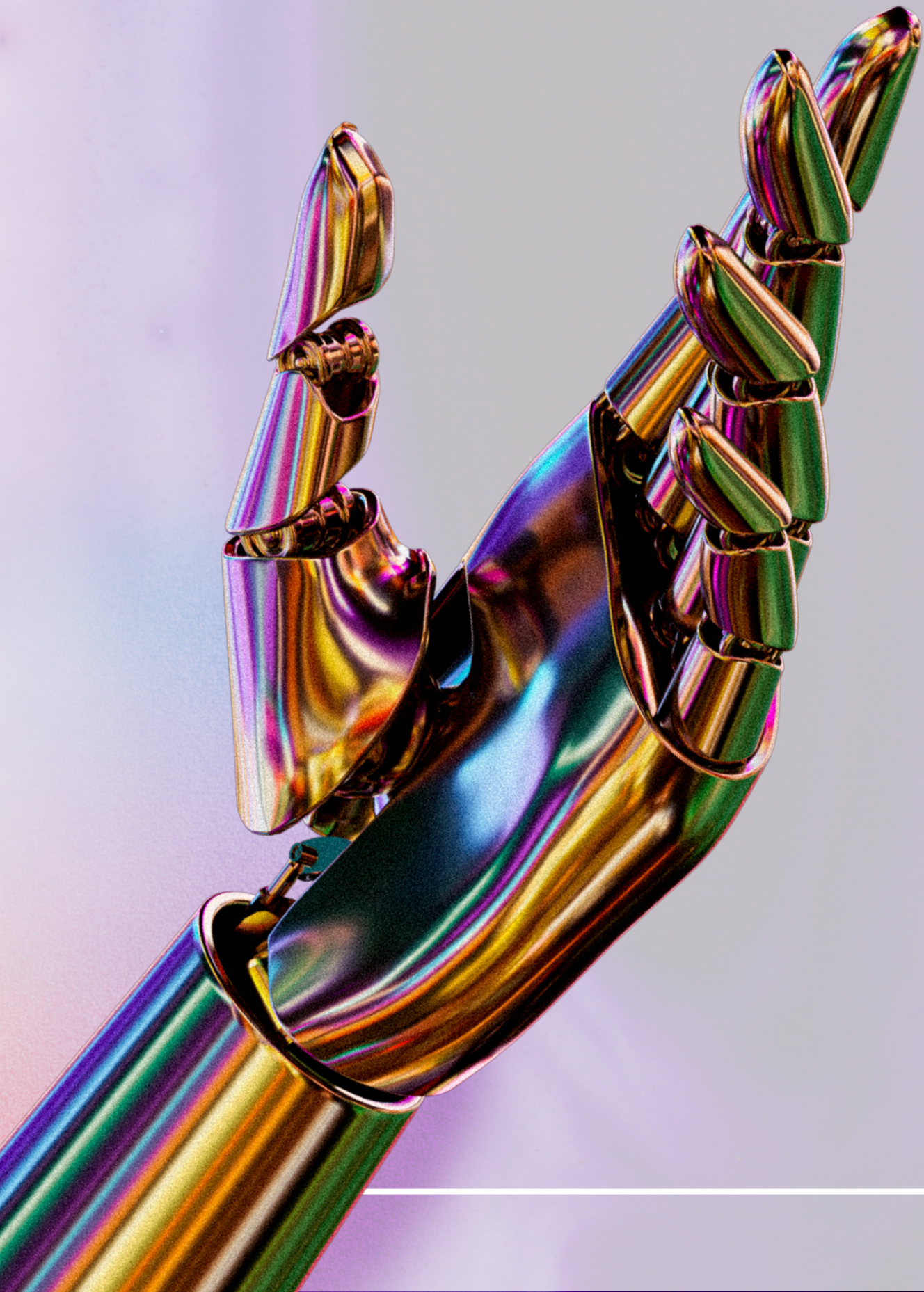
CONVERSATIONAL AI ASSISTANT

- ✓ Interactive chatbot for infrastructure guidance
- ✓ Voice-enabled request submission



SELF-HEALING INFRASTRUCTURE

- ✓ Automatic issue detection and remediation
- ✓ Reduced downtime and operational effort



THANK YOU

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